



OPERATING SYSTEMS

BCS-IV (B)

Project Report

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AZURE VIRTUAL MACHINE

An Ubuntu virtual machine was created on Microsoft Azure. The virtual machine provides a cloud environment for hosting and managing network services. SSH access was enabled for remote administration.

I. FTP SERVER:

- The **VSFTPD** package was installed on the **Ubuntu** Azure Virtual Machine. VSFTPD is a secure FTP service used for file transfer between clients and the server.
- A dedicated **FTP user** account was created. This user is responsible for accessing and managing files on the FTP server.
- The **FTP configuration file** was modified to enable local user authentication, file uploads, and passive mode connections.
- Required **FTP ports** were opened in Azure Network Security Group settings to allow remote access to the FTP server.
- The FTP service was started and verified successfully. The **service status** confirmed that the server was running properly.
- The FTP server was accessed using **Windows File Explorer**. Users were able to browse and manage files stored on the server.
- The FTP server was tested using **FileZilla Client**. A successful connection verified that the FTP server was functioning correctly.

Screenshots:

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (8)						
300	SSH	22	TCP	Any	Any	Allow
301	FTP-20	20	TCP	Any	Any	Allow
302	FTP-PASSIVE	40000-40...	TCP	Any	Any	Allow
303	Allow-HTTP	80	TCP	Any	Any	Allow
310	FTP-21	21	TCP	Any	Any	Allow

```
azureuser@Ubuntu-VM:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1910 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [346 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1666 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [21.1 kB]
```

```
Get:35 http://azure.archive.ubuntu.com/ubuntu noble-security/restricted Translation-en [662 B]
Get:36 http://azure.archive.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:37 http://azure.archive.ubuntu.com/ubuntu noble-security/multiverse Translation-en [7204 B]
Get:38 http://azure.archive.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Fetched 15.9 MB in 2s (7267 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
66 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
azureuser@Ubuntu-VM:~$ sudo apt install vsftpd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ssl-cert
The following NEW packages will be installed:
  ssl-cert vsftpd
0 upgraded, 2 newly installed, 0 to remove and 66 not upgraded.
Need to get 137 kB of archives.
After this operation, 380 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 vsftpd amd64 3.0.5-0ubuntu3.1 [120 kB]
Fetched 137 kB in 0s (416 kB/s)
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl start vsftpd
azureuser@Ubuntu-VM:~$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 10:25:11 UTC; 3min 25s ago
     Main PID: 2188 (vsftpd)
        Tasks: 1 (limit: 973)
      Memory: 700.0K (peak: 1.5M)
         CPU: 9ms
       CGroup: /system.slice/vsftpd.service
               └─2188 /usr/sbin/vsftpd /etc/vsftpd.conf

Apr 21 10:25:11 Ubuntu-VM systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Apr 21 10:25:11 Ubuntu-VM systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

```
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=YES
```

```
write_enable=YES
chroot_local_user=YES
pasv_enable=YES
pasv_min_port=40000
pasv_max_port=40100
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl restart vsftpd
azureuser@Ubuntu-VM:~$
```

```
azureuser@Ubuntu-VM:~$ sudo adduser ftpuser
info: Adding user 'ftpuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'ftpuser' (1001) ...
info: Adding new user 'ftpuser' (1001) with group 'ftpuser (1001)' ...
info: Creating home directory '/home/ftpuser' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for ftpuser
Enter the new value, or press ENTER for the default
  Full Name []: hajrabashir
  Room Number []: 113
  Work Phone []: 03359991170
  Home Phone []: 03710391170
  Other []:
Is the information correct? [Y/n] y
info: Adding new user 'ftpuser' to supplemental / extra groups 'users' ...
info: Adding user 'ftpuser' to group 'users' ...
```

```
azureuser@Ubuntu-VM:~$ sudo mkdir -p /home/ftpuser/ftp
azureuser@Ubuntu-VM:~$ sudo chown nobody:nogroup /home/ftpuser/ftp
azureuser@Ubuntu-VM:~$ sudo chmod a-w /home/ftpuser/ftp
azureuser@Ubuntu-VM:~$ sudo mkdir /home/ftpuser/ftp/files
azureuser@Ubuntu-VM:~$ sudo chown ftpuser:ftpuser /home/ftpuser/ftp/files
azureuser@Ubuntu-VM:~$
```

ftp://135.235.196.37/

Log On As

Either the server does not allow anonymous logins or the e-mail address was not accepted

FTP server: 135.235.196.37

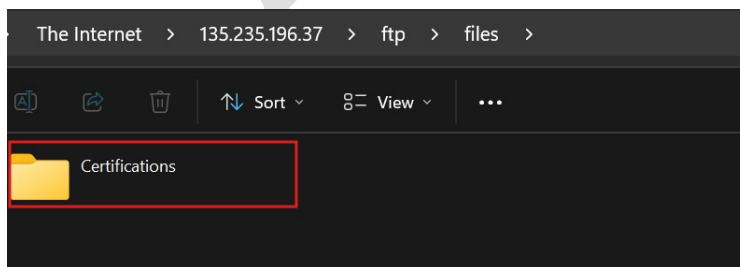
User name:

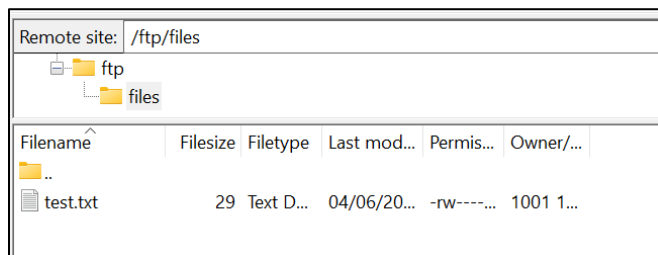
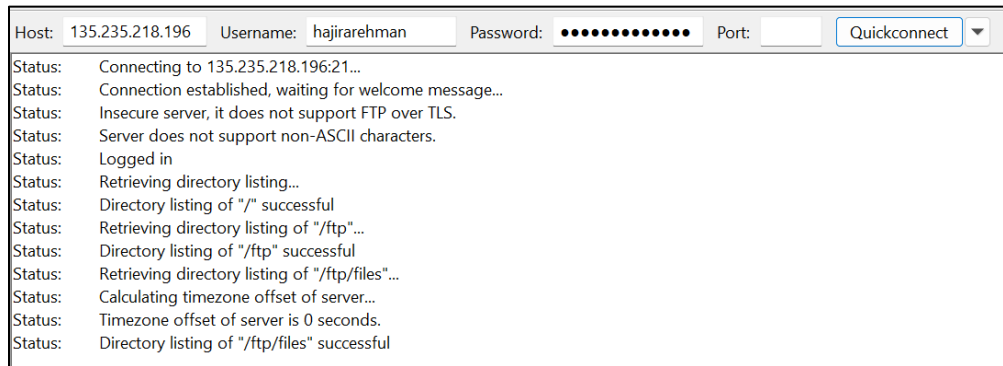
Password:

After you log on, you can add this server to your Favorites and return to it easily.

 FTP does not encrypt or encode passwords or data before sending them to the server. To protect the security of your passwords and data, use WebDAV instead.

☐ Log on anonymously ☐ Save password





II. WEB SERVER APACHE:

- The Apache **HTTP Server package** was installed on the Ubuntu virtual machine to host web content.
- The Apache service was started and verified. The **status output** confirmed that the web server was running successfully.
- **Port 80** was opened in Azure **Network Security Group** settings to allow web traffic from external clients.
- A **custom HTML page** was created and stored in the **Apache web directory** to replace the default web page.
- The web server was tested using a **web browser**. The custom webpage was displayed successfully using the server's public **IP address**.

Screenshots:

```
azureuser@Ubuntu-VM:~$ sudo systemctl start apache2
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl enable apache2  
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-26 19:30:47 UTC; 2min 29s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 15363 (apache2)
    Tasks: 55 (limit: 988)
   Memory: 6.7M (peak: 6.9M)
      CPU: 39ms
   CGroup: /system.slice/apache2.service
           └─15363 /usr/sbin/apache2 -k start
             └─15366 /usr/sbin/apache2 -k start
               └─15367 /usr/sbin/apache2 -k start

May 26 19:30:47 Ubuntu-VM systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 26 19:30:47 Ubuntu-VM systemd[1]: Started apache2.service - The Apache HTTP Server.
```



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.Load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
```

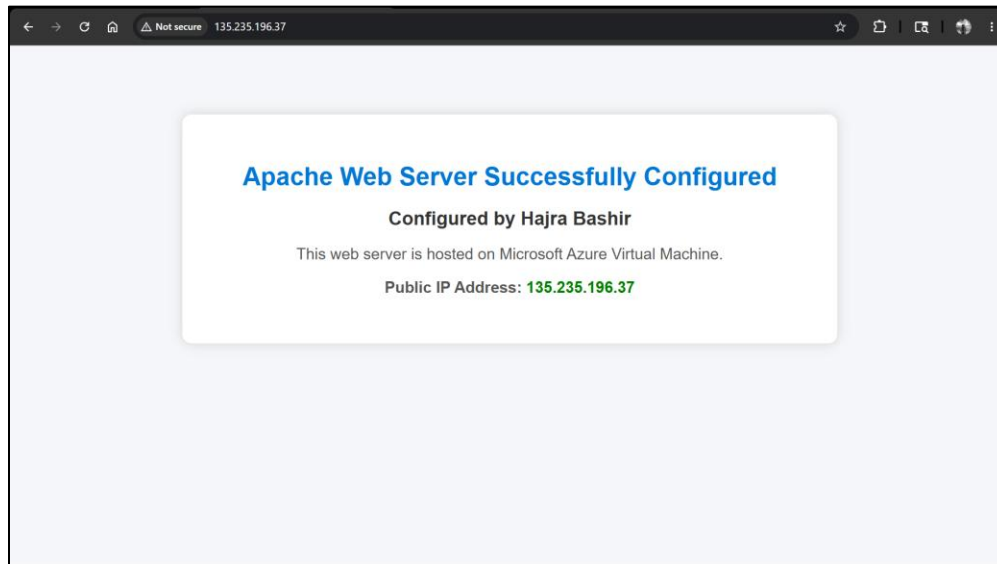
```
azureuser@Ubuntu-VM:~$ cd /var/www/html

azureuser@Ubuntu-VM:/var/www/html$ sudo nano /var/www/html/index.html
```



```
GNU nano 7.2
<html>
<head>
  <title>My Azure Apache Server</title>
  <style>
    body {
      background-color: #f4f6f9;
      font-family: Arial, sans-serif;
      text-align: center;
      padding-top: 80px;
    }
    .container {
      background-color: white;
      width: 60%;
      margin: auto;
      padding: 40px;
      border-radius: 12px;
      box-shadow: 0px 0px 15px lightgray;
    }
    h1 {
      color: #0078D7;
    }
    h2 {
      color: #333;
    }
    p {
      font-size: 20px;
      color: #555;
    }
    .ip {
      color: green;
      font-weight: bold;
    }
  </style>
</head>
```

```
<body>
  <div class="container">
    <h1>Apache Web Server Successfully Configured</h1>
    <h2>Configured by Hajra Bashir</h2>
    <p>
      This web server is hosted on Microsoft Azure Virtual Machine.
    </p>
    <p>
      <strong>Public IP Address:</strong>
      <span class="ip">135.235.196.37</span>
    </p>
  </div>
</body>
</html>
```



III. DNS SERVER:

- The **BIND9 package** was installed to provide DNS services on the Ubuntu server.
- A **custom DNS zone** was created and configured. Domain records were added to **map hostnames** to IP addresses.
- The **DNS service** was started and verified to ensure proper functionality.
- DNS resolution was tested using the **nslookup command**. The successful response confirmed that the DNS server was **working** correctly.

Screenshots:

```
azureuser@Ubuntu-VM:~$ sudo systemctl start named
azureuser@Ubuntu-VM:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-26 19:53:25 UTC; 2min 8s ago
     Docs: man:named(8)
    Main PID: 16299 (named)
      Status: "running"
        Tasks: 8 (limit: 988)
      Memory: 23.0M (peak: 24.8M)
         CPU: 51ms
    CGroup: /system.slice/named.service
            └─16299 /usr/sbin/named -f -u bind
```

```
azureuser@Ubuntu-VM:~$ sudo nano /etc/bind/named.conf.local
```



```
GNU nano 7.2 /etc/bind/name
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "hajra.local" {
    type master;
    file "/etc/bind/db.hajra";
};|
```

```
azureuser@Ubuntu-VM:~$ sudo cp /etc/bind/db.local /etc/bind/db.hajra
```

```
azureuser@Ubuntu-VM:~$ sudo nano /etc/bind/db.hajra
```

```
GNU nano 7.2
$TTL      604800
@ IN SOA hajra.local. root.hajra.local. (
    2
    604800
    86400
    2419200
    604800 )

@      IN  NS      hajra.local.
@      IN  A       135.235.196.37
www    IN  A       135.235.196.37|
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl restart named
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-26 20:09:32 UTC; 10s ago
     Docs: man:named(8)
    Main PID: 16822 (named)
      Status: "running"
        Tasks: 6 (limit: 988)
      Memory: 23.0M (peak: 25.0M)
         CPU: 45ms
    CGroup: /system.slice/named.service
            └─16822 /usr/sbin/named -f -u bind
```

```
azureuser@Ubuntu-VM:~$ nslookup hajra.local 127.0.0.1
Server:          127.0.0.1
Address:         127.0.0.1#53

Name:   hajra.local
Address: 135.235.196.37
```

IV. **DHCP SERVER:**

- The **ISC DHCP Server package** was installed to provide automatic IP address assignments to client devices.
- A **DHCP address pool** was configured by specifying an IP range, subnet mask, gateway, and DNS settings.
- The DHCP service was linked to the appropriate **network interface** to distribute IP addresses on the network.
- The DHCP service was started and verified successfully. The **service status** confirmed proper operation.
- The **DHCP configuration** was tested to ensure that clients could receive network settings automatically.

Screenshots:

```
azureuser@Ubuntu-VM:~$ sudo nano /etc/dhcp/dhcpd.conf
```

```
GNU nano 7.2 /etc/dhcp/dh
# fixed-address fantasia.example.com;
#}

# You can declare a class of clients and then do address allocation
# based on that.  The example below shows a case where all clients
# in a certain class get addresses on the 10.17.224/24 subnet, and all
# other clients get addresses on the 10.0.29/24 subnet.

#class "foo" {
# match if substring (option vendor-class-identifier, 0, 4) = "SUNW";
#}

#shared-network 224-29 {
# subnet 10.17.224.0 netmask 255.255.255.0 {
#   option routers rtr-224.example.org;
# }
# subnet 10.0.29.0 netmask 255.255.255.0 {
#   option routers rtr-29.example.org;
# }
# pool {
#   allow members of "foo";
#   range 10.17.224.10 10.17.224.250;
# }
# pool {
#   deny members of "foo";
#   range 10.0.29.10 10.0.29.230;
# }
#}
subnet 10.0.0.0 netmask 255.255.255.0 {
  range 10.0.0.50 10.0.0.100;
  option routers 10.0.0.1;
  option domain-name-servers 8.8.8.8;
}
```

```
azureuser@Ubuntu-VM:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 60:45:bd:73:88:31 brd ff:ff:ff:ff:ff:ff
    inet 172.16.0.4/24 metric 100 brd 172.16.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::6245:bdf:fe73:8831/64 scope link
        valid_lft forever preferred_lft forever
```

```
azureuser@Ubuntu-VM:~$ sudo nano /etc/default/isc-dhcp-server
```

```
GNU nano 7.2 /etc/default/isc-dhcp-s
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="eth0"
INTERFACESv6=""
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl start isc-dhcp-server
azureuser@Ubuntu-VM:~$ sudo systemctl enable isc-dhcp-server
Synchronizing state of isc-dhcp-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable isc-dhcp-server
```

```
azureuser@Ubuntu-VM:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-26 20:32:07 UTC; 15s ago
     Docs: man:dhcpd(8)
    Main PID: 17503 (dhcpd)
      Tasks: 1 (Limit: 988)
    Memory: 4.0M (peak: 6.0M)
       CPU: 7ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─17503 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf eth0
```

CONCLUSION

This project successfully demonstrated the installation and configuration of FTP Server, Apache Web Server, DNS Server, and DHCP Server on an Ubuntu Virtual Machine hosted on Microsoft Azure. All services were tested and verified successfully, providing practical experience in Linux server administration and cloud-based networking.

REPORT